A fantastic title about minihalos that somehow grips a viewers attention for over 100 pages of detail

by

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Abstract

A truly brilliant abstract that captures the nuance of what I have done, without being overly detailed and dense so that a reader throws it out the window.

Acknowledgments

People who helped me during my Thesis: Yvette, CASA people, sibling.

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Chapter 1

Introduction

When looking up into the sky on a clear night, those without an interest in astrophysics might imagine many of the points in the sky as galaxies just like our own, floating alone in the deep void of space. Although true to an extent, it doesn't paint an accurate picture of the cosmos. In reality, many of these form galaxy clusters, with tens to thousands of individual galaxies becoming gravitationally bound into a single massive structure (somewhat analagous to the planets in our solar system, but on a vastly different scale).[1] Galaxy clusters are the largest gravitationally bound structures in the universe, with total masses of order $10^{14}-10^{15}$ $M_\odot$ and diameters larger than 1.5 million parsec.[2]

One might naturally expect that the majority of the cluster mass to be from the innumerable galaxies held within, however, only a few percent of the total mass comes from this baryonic matter (such as stars, planets, etc). The remainder of the baryonic mass comes from a hot diffuse gas (10^7-10^8 K) spread throughout the cluster called the intracluster medium (ICM). This dilute gas ($\sim 10^3$ particles m^{-3}) is held together by the gravitational pull of the cluster, with the highest density in the core and sparsest at the edges (Meekins et al., 1971; Serlemitsos et al., 1977). Even then, this amount of visible baryonic matter only comprises 15% of the gravitational mass calculated from the dynamics of the cluster, a huge discrepancy that was a key piece of evidence for the existence of dark matter. Indeed, the rough consensus from decades of study on galaxy clusters has $\sim 3\%$ of the mass budget taken by galaxies, $\sim 12\%$ from the ICM, and a staggering $\sim 85\%$ from unobservable dark matter (Blumenthal et al., 1984; Vikhlinin et al., 2006; Pratt et al., 2019).

Between these clusters, long filaments of galaxies span across space, with their own warm-hot intergalactic medium (WHIM) mimicking the role of the ICM. The WHIM is cooler and more diffuse than the ICM, with densities $\leq 10^2$ particles m^{-3} and temperatures 10^5-10^7 K. These

[1] Smaller clusters with only tens of galaxies are often known as groups.

[2] For those unaccustomed to astrophysical units, $M_\odot$ is the solar mass and defined as $1\ M_\odot \approx 10^{30}$ kg. A parsec is defined as $1\ \text{pc} \approx 3\times10^{16}$ m. These unusual units are far more suitable at the scale of the universe.

filaments are larger than clusters, though not gravitationally bound, and create a cosmic web in the sky with galaxy clusters located at the intersections. This cosmic web is the largest structure in the universe, a vast network of dark matter that we can trace via the observable galaxies and clusters (Bond et al., 1996; Pimbblet, 2005) with ongoing work in detecting these filaments observationally (Tudorache et al., 2025). Clusters are known to form in part due to accretion of the WHIM these at intersections of the cosmic web, as well as by mergers between smaller clusters (Press and Schechter, 1974; Voit, 2005; Kravtsov and Borgani, 2012).

An important distinction should be made when analysing clusters, between "disturbed" clusters undergoing merger events or still in a turbulent state from a recent merger, and "relaxed" clusters which have settled into an undisturbed state. Relaxed clusters favour theoretical studies due to their greater spherical symmetries, fewer substructures, and therefore, simpler modelling (Mantz et al., 2016). Many relaxed clusters have also been found to feature a "cool core", where the temperature at the centre of the cluster is significantly lower than the region outside the core. The high density of the ICM in the core causes a large outflux of X-ray emission, great enough to lower the cooling time below the expansion rate of the universe (known as Hubble time) . In a closed system, this develops into a cooling flow causing the core temperature to drop and an inward gas flow (Peterson and Fabian, 2006; Hudson et al., 2010). Observational evidence supports this theory, however, the magnitude of observed cold gas is much lower than predicted, implying the existence of a heating mechanism to counteract the cooling flow. One such source are active galactic nuclei (AGN), supermassive black holes at the centre of some galaxies that are consuming massive amounts of matter, and producing radio synchrotron emission. This emission creates jets and lobes that have been observed interacting with the ICM, distributing energy in "radio-mode" feedback (Fabian, 2012). The exact method of this heating is still disputed and undergoing research.

This can be stated better I think.

Within the high mass subset of relaxed cool core clusters ($M_{500} > 6 \times 10^{14}$ M$_\odot$)[3] recent works have shown that 80% contain radio minihalos (Giacintucci et al., 2017). Less common in lower mass clusters, and unobserved within non-cool core clusters, minihalos are diffuse radio sources with currently unknown origins. This radio emission is localised within the cool core, with radii from 50 kpc to $0.2R_{500}$.[4] This emission cannot be directly formed by radio sources such as AGN, as there is no observed spatial or spectral connection at any resolution or frequencies studied thus far. Furthermore, the spatial extent of these halos is well beyond the expected extent for particles sourced from an AGN (e.g. Bagchi et al., 2002; Giacintucci et al., 2014). Understanding the origins and physics behind minihalos is an ongoing process, with viable production methods proposed, but a lack of conclusive observational evidence to differentiate between them. This

[3]The value M_{500} refers to the total mass within a radius of R_{500}, defined by the distance at which the clusters average enclosed density is 500 times the critical density of the universe at the cluster redshift. The critical density is a cosmological quantity that has varied over the lifetime of the universe.

[4]The radius $0.2R_{500}$ is a standard metric to separate the core and the outer ICM. Within this radius, the cooling, AGN feedback, and similar processes become significant to the behaviour of the ICM.

thesis will provide new observational data and a generalised process to provide further evidence to differentiate between the proposed processes.

1.1 Observations of Galaxy Clusters

The question that should first be asked is, how *do* we know all of this information about galaxy clusters? How are we able to measure the temperature, density, pressures within these clusters without travelling the impossibly vast distances to get there? The answer to this question is simple, by observing the radiation emitted from these structures. Different processes within the cluster, and more specifically ICM, provide information about the physical environment. This radiation is emitted across the electromagnetic spectrum and we design telescopes to capture and relay different bands in order to identify different components of what we observe. The question then becomes, how do observations at different frequency bands inform us to the structure and processes occuring within a cluster?

1.1.1 Optical

The vast majority of optical light from clusters comes from stars within the component galaxies. Although incredibly informing about the stars and galaxies themselves, in the context of minihalos, optical light has limited use. Optical observations are useful in determining the component galaxies of a cluster, alongside investigating the radial velocities of galaxies. Since the radial velocity is dependent on the gravitational potential of the cluster, the distribution allows us to estimate the clusters mass (Voit, 2005). This analysis is difficult for more distant clusters and disturbed clusters, as the dynamical state will affect the radial velocities and skew the resulting mass.

1.1.2 X-ray

The hot gas of the intracluster medium forms a thin plasma, which has a degree of ionisation that depends on the temperature and density. We observe an X-ray spectrum from this gas with both continuum and line characteristics, and photon energies between 0.1 keV and 100 keV.[5] The energy of the photons is dependent on the temperature of the ICM, resulting in most emissions being in the narrower range of 1 to 10 keV (Schneider, 2014).

Line emission occurs due to electron transitions between orbitals within an atom. The temperature and density of the ICM changes the measured rate of this line emission, however, this becomes limited at higher temperatures as the ionisation fraction increases. The distribution of line emissions can also be an indicator of abundance of elements such as C, N Ne, Mg, Si,

[5] X-ray emission is historically measured in kilo-electron volts (energy) rather than frequency or wavelength.

S, Ar, Ca, and Fe. This is particularly evident at lower temperatures ($\lesssim 2$ keV) where the total emissions can be dominated by line emission (Voit, 2005; Schneider, 2014).

Continuum radiation is due to brehmsstrahlung, or free-free radiation, which is where electrons are deflected via Coulomb interaction with nearby nuclei. This deflection causes an acceleration induces a change in kinetic energy which is emitted as a photon. The spectral energy distribution of these emissions is related to the kinetic energy of the particles, and therefore, the temperature of the ICM. The scale of emissions is similarly impacted by the number of particles, and thereby the density of the ICM ρ_g in a given region. This gives rise to the following relationship to total continuum emissivity (Schneider, 2014).

$$\epsilon_c \propto T^{\frac{1}{2}} \rho_g^2 \tag{1.1}$$

X-ray spectroscopy is a powerful tool to characterise the ICM and a clusters dynamical state. In general, telescopes trade off between high spatial or spectral resolutions, i.e. between detecting exactly where a photon originated from or an accurate measurement of the photon energy. This becomes somewhat of a constraint when observing galaxy clusters as we require reasonable spatial and spectral resolutions for these large objects. One common instrument is the space telescope *Chandra X-ray Observatory* launched in 1999 . Many studies of minihalos use *Chandra* observations to map the structure of the cool core and investigate potential relationships to the radio emission (e.g. Giacintucci et al., 2024). However, due to the aging detectors and contaminant build up on *Chandra*, the sensitivity of the telescope is decreasing. This is particularly bad at low energies (< 2 keV), where sensitivity is below 10% of where it was two decades ago (Grant et al., 2024). A new telescope *XRISM* was launched in late 2023 and was designed as a successor to *Chandra*. Over the last year researchers have begun to utilise the greater resolution to more deeply probe the dynamics of the ICM in some clusters (Sarkar et al., 2025; XRISM Collaboration et al., 2025). Although not the focus of this thesis, X-ray observations provide key ancillary data in investigating the processes involved.

Cite/footnote telescopes

1.1.3 Radio

Radio observations cover several orders of magnitude, with frequencies from 10 MHz to over 100 GHz. This massive range comes with several difficulties for astronomers, with the primary one being the angular resolution of the telescope. Angular resolution θ defines the smallest area that can be resolved given an aperture and the observing wavelength.[6] Equation 1.2 gives the formula for angular resolution in arcseconds, or $\frac{1}{3600}$th of a degree, for an aperture of diameter D and observing wavelength λ.[7]

$$\theta = 2.5 \times 10^5 \frac{\lambda}{D} \tag{1.2}$$

[6]The angular resolution of a human eye (with 20/20 vision) is 1 arcminute. *See note 7*

[7]Arcseconds (”), arcminutes ($1' = 60$”), and degrees are common units in astrophysics as declination, the celestial equivalent to latitude, is measured in these units.

Radio waves range from 3 cm to 3 km, orders of magnitude larger than optical ($\sim 1\mu$m) and X-ray (0.01−10 nm). It is then clear that to resolve the same object at radio frequencies as X-ray or optical, vastly larger telescopes would be necessary. In order to observe over this range several types of telescopes have been developed, including single dish telescopes, interferometer arrays, and very long baseline interferometry (VLBI).

The size of a single dish antenna is limited by both practical construction and from scientific usability. The largest dish is the *Five-hundred-meter Aperture Spherical Telescope* (FAST) located in Guizhou China, though of the 500 metre diameter only a 300 metre area is usable at a time. Furthermore, as an immovable dish, it is limited in its view of the sky, with a maximum angle of 40 degrees from zenith. Several smaller 100 metre diameter dishes exist such as the *Green Bank Telescope* in West Virginia United States, and the *Effelsburg Telescope* near Bonn, Germany. These dishes also have a limited view of the sky and variable amounts of steerability due to their size, alongside hard limits on their resolution at higher radio frequencies. These restrictions developed into interferometry, the use of an array of single dish telescopes in tandem to simulate a larger dish. The *Karl G. Jansky Very Large Array* (VLA) in New Mexico United States, is comprised of 28 fully steerable 25-metre radio telescopes mounted on railway tracks in a Y-shaped array, observing between 73 MHz and 50 GHz. The *MeerKAT* telescope in the Meerkat National Park of South Australia contains 64 13.5-metre telescopes and operates between 1000 MHz and 10 GHz. These arrays can simulate observations by dishes several kilometres in diameter, while being able to observe to low sky elevations. VLBI takes this concept even further by using radio telescopes across different continents, synchronised by atomic clocks, to simulate dishes comparable to the size of the Earth. This thesis uses observations taken by the VLA, and the physics of these telescopes will be discussed in more detail in Chapter 2.

Cite/footnote telescopes

Radio emission detected by these telescopes falls into the category of 'synchrotron' emission. This synchrotron emission is produced by particles travelling at relativistic velocities through a magnetic field, emitting photons at radio energies characterised by their velocity and the magnetic field strength (Feretti et al., 2012; Schneider, 2014). This is due to the Lorentz force ($\vec{F} = \vec{v} \times \vec{B}$) perpendicular to their direction of motion, which induces a helical path through the field, and therefore constant acceleration. This is analagous to brehmsstrahlung radiation, but occurs due to a magnetic interaction instead electric, which is why it is somewhat rarely called 'magnetobrehmsstralung'. A detailed discussion of the spectral distribution and features of this radiation is given in Chapter 2.

1.1.4 Gamma

The last region of interest for minihalo research is in detecting gamma radiation, or gamma-rays. This is the high energy limit for astronomy, where telescopes are designed to detect photons with

energies in the MeV and GeV range.[8] These prove difficult to not only detect, due to the majority of rays being absorbed in the atmosphere, but also to accurately describe their origin (Schneider, 2014). The *Fermi Gamma-ray Space Telescope* was built specifically for this research and was launched in 2008 carrying the Large Area Telescope (LAT), with an energy range between 30 MeV and 300 GeV.

There are various sources for gamma-rays, such as Gamma Ray Bursts (GRB), which are extremely short and intense flashes. Many of these have been spatially located to coincide with bright AGNS called blazars, and rapidly spinning magnetised neutron stars called pulsars. Another source is from gamma-ray background radiation, which is expected be via neutral pion decays resulting from proton-proton collisions. Attempts to detect gamma-ray emissions connected to a clusters ICM have been ongoing as they allow us to put limits on effectiveness of proposed mechanisms to create radio halos (e.g. Reimer et al., 2003; Ackermann et al., 2010; Zandanel and Ando, 2014). Though progress has been made for giant radio halos, in particular firm constraints for the Coma cluster (Brunetti et al., 2012), minihalos have a smaller spatial extent and are much fainter, with no conclusive studies. The role of gamma rays in determining the processes creating minihalos is detailed in Chapter 2.

1.2 Diffuse Radio Emission

Add a picture for a minihalo. Maybe Perseus from Fig 3 van Weeren et al. (2019) or own.

It has long been known that some clusters exhibit diffuse radio synchrotron emission that cannot be directly associated with radio galaxies within the cluster. This synchrotron emission can take many spatial arrangements, with the primary categories being radio halos, radio relics, and revived AGN sources. Radio halos are extended sources that trace the ICM distribution and operate at the scales of giant radio halos, matching the full ICM extent, and minihalos that are contained within a clusters cool core. Relics instead reflect particles that have been reaccelerated to relativistic energies by shock waves travelling through the ICM, such as the violent energies released during merger events. Revived AGN sources are radio plasma that has been emitted from and AGN and re-energised similarly to the radio relics.

1.2.1 Production

A key question in relation to the radio halos is how they are produced. The relativistic particles producing this emission have a characteristic lifetime due to the energy losses as they travel through space, and therefore a limited diffusion length-scale in which we should be able to observe them. Within a weak magnetic field ($\sim 10^{-7}$ G) these distances are about 10 pc, which even accounting for bulk motion of the ICM and more favourable diffusion environments are well below the Mpc scales observed in many clusters (Bagchi et al., 2002). Two primary theories

[8] Like X-ray, gamma-rays are measured by their energy rather than frequency.

have been brought forward to explain this phenomenon, one being turbulence in the ICM causing "reacceleration" of once relativistic particles (e.g. Brunetti et al., 2001; Gitti et al., 2002; ZuHone et al., 2013; Zandanel et al., 2014). The other is inelastic collisions between relativistic cosmic ray protons and thermal protons, termed "hadronic" or secondary production (e.g. Dennison, 1980; Dolag and Enßlin, 2000; Keshet and Loeb, 2010; ZuHone et al., 2015; Ignesti et al., 2020).

Both of these mechanisms produce similar observable features, which makes disentangling the true source much more challenging. Investigation into spatial extents and spectral features of giant radio halos has given the reacceleration theory more credence than hadronic production (Brunetti et al., 2012), however, the debate is still ongoing for minihalos. A detailed explanation of these processes is provided in Section 2.3. Due to their faint nature and smaller size, observations require high sensitivity and resolution in order to accurately capture an image.

1.2.2 Minihalo detections

Much effort has been devoted towards detection and validation of minihalos in order to build a comprehensive sample. Over the last two decades the number of detected minihalos has increased substantially, from 10 minihalos and minihalo candidates in 2011 (Feretti et al., 2012), to 22 in 2017 (Giacintucci et al., 2014), and 40 in 2024 (Giacintucci et al., 2024). A recent sample of 23 confirmed minihalos is detailed by Giacintucci et al. (2019), though several more have been confirmed since (e.g. Richard-Laferrière et al., 2020; Kale et al., 2022; Lusetti et al., 2024).

One of the major issues with detection is in separating the minihalo emission from the much brighter sources embedded within and nearby it. Alongside the minihalo in the core is the brightest cluster galaxy (BCG), the most radio-loud AGN within the cluster. Minihalos typically have a surface brightness of $1 \sim 5$ μJy arcsec^{-2}, while the BCG can have a flux 1000 times higher.[9] The BCG must then be masked or subtracted away from an observation to reveal the minihalo, but this requires the observation to have good resolution of point sources as well as the larger spatial minihalos. This requirement makes current general sky surveys insufficient for detecting minihalos, with some studies suggesting that next generation radio telescopes could reveal thousands more minihalos due to improved capabilities (Gitti et al., 2018).

1.3 This Thesis

1.3.1 Aims

The primary goal of this thesis is to calibrate, image, and analyse new radio observations of 12 minihalos at 10 GHz. These observations were performed by the VLA in C and D configuration

[9]The Jansky (Jy) is a unit measurement of spectral flux density common in radio astronomy. It measures the rate of energy received per square meter per hertz, or 10^{-26}W m^{-2} Hz^{-1} in SI units.

over two periods in early 2024 and early 2025. This data constitutes the first broad survey of minihalos at high radio frequencies.

This data will complement previous surveys at lower frequencies taken by the *Low-Frequency Array* (LOFAR) at 140 MHz and MeerKAT at 1.3 GHz (Riseley et al., 2022). Processing this data will allow measurements of the minihalo flux to investigate if steepening of the spectrum occurs at higher frequency, which will assist in differentiating between the reacceleration and hadronic scenarios.

1.3.2 Work

In pursuing these aims, I downloaded the raw interferometric data from the National Radio Astronomy Observatory (NRAO) Archive[10] and documented the calibration and imaging process. I then performed various analyses on the final images for each minihalo to evaluate the limits and usefulness of 10 GHz observations for characterising the production mechanisms of minihalos.

This work was done in a source controlled repository[11] so that the results are reproducible and documented. Due to the size of the datasets, (uncalibrated ~ 1.4 TB compressed, calibrated ~ 2.5 TB, imaging ~? TB), this is stored on the Storage fOr Learning And Research (SOLAR) large scale storage provided by Victoria University of Wellington.

Add final size

The repository is split into two primary sections, *Data Processing* and *Image Processing*. The first section follows the calibration of the raw data and separation from observation runs into cluster specific datasets, split by observation id. The following image processing includes the cleaning and creation of the images from the trimmed datasets and the associated analysis and is split by cluster.

1.3.3 Layout

The introduction gives a brief overview for important concepts relating to minihalo physics and research.

Chapter 2 details previous research on minihalos and deeper explanation of the hadronic and turbulent reacceleration processes.

Chapter 3 introduces the collected data and explains the calibration and imaging process used.

Chapter 4 explains the tools and techniques used in the analysis of the final images.

Chapter 5 details the results found for each cluster, with the context of previous studies.

Chapter 6

Chapter 7

[10] https://data.nrao.edu/portal/

[11] https://github.com/MoonlitMoo/Masters-2025

Chapter 2

Radio Halos

2.1 Synchrotron emission

Since the first discoveries of diffuse radio emission, much research has been devoted towards understanding the mechanisms behind the production of synchrotron electrons. These electrons are also known as 'cosmic ray' electrons (CRe) or non-thermal electrons to differentiate them from the thermal electrons of the ICM. The CRe have an individual energy $E = \gamma m_e c^2$ dependent on their Lorentz factor γ.

$$\gamma = \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{2}} \tag{2.1}$$

In the case of radio halos we can expect $\gamma \gtrsim 10^4$ which corresponds to velocities within 10^{-4}% of the speed of light. Throughout their radiative lifetime CRe produce radio synchrotron emission with energy as a function of the perpendicular magnetic field and their Lorentz factor. This is broadly a continuum radiation with the synchrotron power described by the following equation (Feretti and Giovannini, 2008).

$$-\frac{dE}{dt} = \frac{2e^4}{3m_e^4 c^7}(B\sin\theta)^2 E^2 \tag{2.2}$$

They also interact with thermal protons within the ICM in a process called inverse Compton (IC) scattering.[1] The scattered photon has a boosted energy that is proportional to its orginal energy. In units of frequency this relation is:

$$\nu_p = \frac{3}{4}\gamma^2 \nu_{p,0} \tag{2.3}$$

For CRe under consideration this results in a factor $\sim 10^6$. The CMB is a source of thermal photons throughout the cluster with a temperature of T = 2.73K, to which the Planck function

[1] Standard Compton scattering has ultrarelativistic photons interacting with thermal electrons, the opposite to what we are considering.

peaks at $\nu \approx 1.6 \times 10^{11}$ Hz. Which results in high energy X-ray emission (Sarazin, 1999; Govoni and Feretti, 2004).

Synchrotron radiation is insufficient to directly measure the magnetic field, and the standard method is to instead measure the equipartition field B_{eq}. This field is found by minimising the total energy of the magnetic field potential and that of the relativistic particles. The IC emission is independent of the magnetic field however, meaning that comparisons between these values would allow direct measurement of the magnetic field strength (Govoni and Feretti, 2004; Wik et al., 2014). Unfortunately, detection of the X-ray IC component is difficult due to low X-ray energies dominanted by the thermal component, while upper energies have lacking spatial resolution in current instruments. An interesting topic in its own right, IC emission and cluster magnetic fields are outside the scope of this thesis.

2.2 Lifetime of synchrotron emission

The energy losses of the CRe as they interact with the electromagnetic fields and particles in the ICM eventually result in them no longer being energetic enough to produce radio emission. Considering losses due to synchrotron and IC processes, the lifetime of a CRe can be given as the following function of the magnetic field in the ICM B and the magnetic field of the CMB B_{CMB} where the magnetic fields are measured in μG.

$$\tau = 32 \frac{B^{\frac{1}{2}}}{B^2 + B_{CMB}^2} \left[(1+z)\nu\right]^{-\frac{1}{2}} \tag{2.4}$$

We can also define $B_{CMB} = 3.25(1+z)^2$ μG to account for the redshift of the clusters in question. As the majority of observed minihalos have redshifts in the range $[0.1, 0.3]$[2] and magnetic fields within the ICM are estimated at $\lesssim \mu$G, the maximum expected lifetime of a CRe is about 10^8 years (Sarazin, 1999; Feretti et al., 2012; Gitti et al., 2018).

Then we can find the expected extent for these particles, in absentia of external forces, as $\sim 10^{16}$ m or 1 pc. This is orders of magnitude below the extents of observed diffuse radio emission, so we then consider that the CRe are moving within the bulk motion and magnetic fields of the ICM. If we assume that there is a strong of turbulence to promote diffusion of CRe, we can use the Bohm approximation $D = \frac{k_B T}{eB}$, which models the diffusion of plasma across a magnetic field. Bagchi et al. (2002) finds that The expected length-scale of this diffusion (at $z = 0.3$) is that of a random walk across its lifetime with a diffusion coefficient κ and length-scale

[2] A handful minihalos are notably outside this range, with the Phoenix cluster at $z = 0.596$, ACTJ0022-0036 at $z = 0.8$, and a recent candidate SpARCS1049+56 at $z = 1.709$ (HLavacek-Larrondo et al., 2025)

L_{diff},[3]

$$\kappa = 3.31 \times 10^{21} \left(\frac{E}{\mathrm{GeV}}\right)\left(\frac{B}{10^{-1}\mu\mathrm{G}}\right) \mathrm{m\ s^{-1}} \tag{2.5}$$

$$L_{diff} = \sqrt{\kappa\tau} = 6.7 \left(\frac{B}{10^{-1}\mu\mathrm{G}}\right)^{-\frac{1}{2}} \mathrm{pc} \tag{2.6}$$

Once again, we find that the expected diffusion and halo produced by a point source is measured in the 10s of parsecs, far below the kiloparsec and megaparsec halos observed. Therefore, we must assume that these particles are being reaccelerated or produced *in-situ* by a process that is extended over the kiloparsec scale and long-lived across the particle lifetime.

Before discussing the proposed processes, it seems prudent to first mention the standard assumptions regarding the population of relativistic CRe within the ICM. These are that it is homogeneous, isotropic, and follows a power-law energy distribution $n(E) \propto E^{-\delta}$. This then gives us the following basic observables for a population of initially relativistic particles (see van Weeren et al. (2019), Feretti et al. (2012) and references within).

1. The synchrotron emissivity is dependent on the number density of the CR electrons and the strength of the magnetic field. As such, the emissivity also follows a power-law $S(\nu) \propto \nu^{-\alpha}$ which is related to δ by $\alpha = (\delta - 1)/2$. Typical minihalo spectral indices are steep with $\alpha > 0.7$.

2. As the emitting particles lose energy the particle energy distribution changes, and therefore, the emitted radio spectrum changes. This appears as a steepening or curved spectrum at high frequencies, rather than a flat power-law. The break frequency at which this occurs is related to the time of acceleration. This means that radio events that have stopped will show an aged spectrum with a characteristic break, such as radio relics, while ongoing events are continually injecting new particles and don't feature the same steepening due to age.

3. Radiation from CR electrons passing through a uniform magnetic field is linearly polarised. Non-uniform and disordered fields reduce the linear polarisation.

Several models have been developed to describe the spectral shape, depending on the initial injection spectrum α_{inj}, break frequency, and any continuous injection after the initial state. The Jaffe-Perola (JP; Jaffe and Perola 1973) model assumes all CRe are created in a single burst, before aging with continuous scattering of electron pitch angles. This predicts that the pitch angle (between the electron velocity vector and local magnetic field vector) will be isotropic due to scattering. The Kardashev-Pacholczyk (KP; Kardashev 1962, Pacholczyk 1970) model is similar to the JP model as it uses the same initial conditions, but evolves without pitch angle

[3]The length-scale diffusion is calculated as a random walk, scaling with the square root of time, rather than linearly.

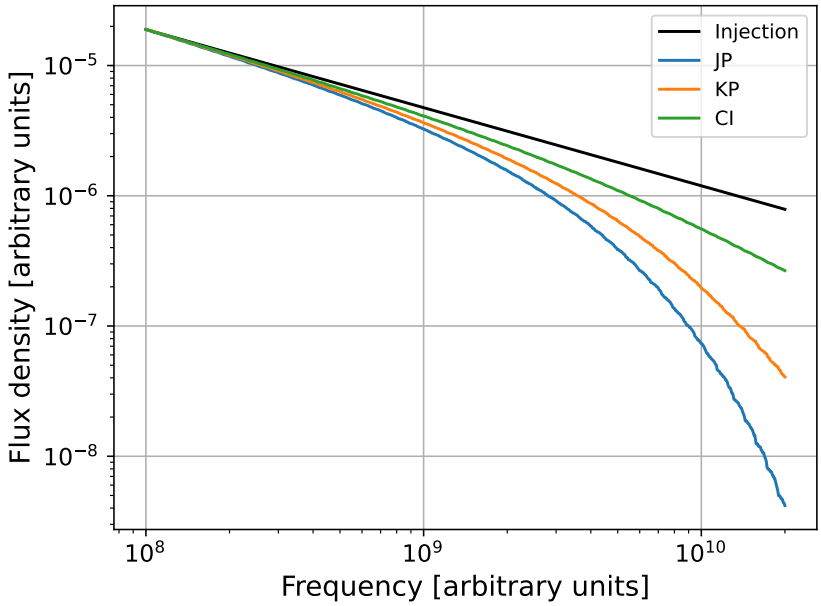

Figure 2.1: A representation of radio spectral shapes with $\alpha_{inj} = -0.6$. The injection spectrum depicts the shape prior to energy losses.

scattering. Continuous injection (CI) models can describe sources that existed for longer than a single burst, such as until current time (CI-on; Kardashev 1962), or for a fixed amount of time τ_{on} (CI-off; Komissarov and Gubanov 1994). An example of how these models differ in their shape is provided in Figure 2.1

2.3 Production mechanisms

With this understanding of the spectrum of CRe, we now consider the methods that may produce them. This can be broken down into primary models involving first and second order Fermi acceleration, and secondary models where the electrons are a byproduct or result from decay.

First order Fermi acceleration (Fermi-I) is also known as diffuse shock acceleration (DSA). This occurs when particles cross back and forward over the shock front, as they are reflected by magnetic instabilities upstream and downstream from the shock. Each crossing of the shock front increases the particle energy, raising it to relativistic levels.

2.3.1 Turbulent reacceleration

This process takes a pre-existing lower energy CRe population of $\gamma \sim 100$ and boosts it to the factors observed $\gamma \gtrsim 1000$. This occurs via interactions with plasma waves and shocks within the ICM which can be caused by several factors, such as violent cluster mergers and gas sloshing within the core (Brunetti et al., 2001; Fujita et al., 2004). This environment naturally allows

these low energy 'seed' electrons to undergo second order Fermi (Fermi-II) acceleration. This type of acceleration is stochastic as particles are scattered by the magnetic instabilities from the randomly moving plasma waves and has an efficiency scaling. Particles meeting these waves head-on are reflected with an increased energy, while particles in a head-tail collision are reflected with a lower energy. The chance for a head-on collision is higher than that of a head-tail collision allowing for a net gain of energy. The mean energy gained is proportional to the speed of the waves and energy of the particle via,

$$\Delta E \propto \frac{v_{wave}^2}{c^2} E \tag{2.7}$$

The random nature of Fermi-II acceleration makes for an inefficient process. Brunetti and Lazarian (2007) show that reacceleration of relativistic particles moving through magnetohydrodynamic (MHD) turbulence are primarily accelerated by two Fermi-II type mechanisms.

The first mechanism is the damping of fast magnetosonic waves due to resonant transit time damping (TTD). The compressible and low-frequency plasma waves can resonate with gyrating particles at the zeroth order, $\omega = v_{\|} k_{\|}$, where ω is the frequency of the wave, $k_{\|}$ is the wavenumber, and $v_{\|}$ is the speed of the particle along the axis of the magnetic field. The random accelerations can then be captured as a random walk in momentum space, giving rise to a momentum-diffusion coefficient $D_{pp,TTD}$. ZuHone et al. (2013) simulate this process numerically in the case of minihalos and simplify this diffusion coefficient to be proportional to the average wavenumber $\langle k \rangle$, turbulent velocity of the particle v_t, and fraction of turbulent energy in compressible motion R_c.

$$D_{pp,TTD} \propto \langle k \rangle v_t^2 R^c p^2 \tag{2.8}$$

The second mechanism is via non-resonant compression and expansion. The adiabatic compression increases the particle momentum, while expansion reduces it. Within a Fermi-II process, compression is statistically more likely and so a net acceleration is acheived. Here ZuHone et al. (2013) take the fast diffusion limit, where particles escape the eddies of compression before they re-expand. This non-resonant momentum-diffusion coefficient scales as

$$D_{pp,C} \propto \frac{V_l^2}{D} p^2 \tag{2.9}$$

where V_l is the compressive velocity at scale l, and $D \propto c l_{mfp}$ is the spatial diffusion rate for a mean free path of l_{mfp}.

The reacceleration efficiency is then given via $\chi = \frac{4D_{pp}}{p^2}$, where D_{pp} is the sum of the prior diffusion coefficients. The total rate of momentum increase for each particle can then be clearly written as the sum of two Fermi-II processes when compared to Equation 2.7.

$$\frac{dp}{dt} = \chi p \tag{2.10}$$

$$\frac{dp}{dt} \propto \langle k \rangle R^c v_t^2 p + \frac{1}{D} V_l^2 p \tag{2.11}$$

One remaining loose thread in this explanation is the source of the seed electrons themselves. A possible explanation is that the seed population is provided from radio AGN and spread throughout the cluster core by the gas sloshing (Brunetti et al., 2001). The CRe from the central BCG lose energy and fizzle out of our detectable range by the physics described earlier, however, they don't stop existing after this timescale. These 'relic' CRe are lower energy but still relativistic with $\gamma \sim 100$, radiating at radio frequencies that we are unable to detect. In this state they still continue to move and disperse through the core, creating the seeds for the large scale minihalos that we observe. The central BCG may not be the only source, other AGN within the cluster, galactic winds, supernovae, may all contribute relic CRe to be later reaccelerated. ZuHone et al. (2013) find that the sloshing should be sufficient to redistribute these particles to create a minihalo. Hadronic interactions as described in a later section could also contribute seed electrons.

2.3.2 Observables of turbulent reacceleration

With this understanding we can now explore the spectral and spatial features that we expect to observe from emission resulting from the reacceleration scenario. Turbulence is an inherently transient and intermittent phenomena. This spatial variance means that when we sample a given region of space, we may instead be observing several 3D components superimposed on our 2D image. Areas of with higher turbulent velocity inspire greater magnetic fields, and greater levels of emission (ZuHone et al., 2013). The lack of a steady state and variable components means that we might expect variation in the flux brightness and spectral index across different locations of the minihalo (Riseley et al., 2022). However, due to the integration over the line of sight, these artefacts could be averaged and less visible in the 2D observation.

The particles must also trace the distribution of turbulence within the ICM, leading to a spatial bounding of the minihalo within the gas sloshing of the core. The boundary of the cluster core is often associated with cold fronts visible in X-ray data, allowing comparison between X-ray and radio images. ZuHone et al. (2013) find that simulated turbulent re-acceleration is indeed boundes by the cold fronts. They also find that depending on the projection through the ICM, it may be difficult to locate the cold fronts in X-ray images due to the afore mentioned line of sight integration .

Image from ZuHone et al. (2013) of boundary?

A spectral feature of the reacceleration model is due to the balancing of reacceleration and losses, creating a cutoff energy at which the population of CRe declines more sharply. This is indeed the logic behind the CI model visualised in Figure 2.1, with a steepening in the spectral index beyond the break frequency. ZuHone et al. (2013) note that as there is a range of magnetic field strengths within the core, this break frequency will change between CRe, causing a smearing of the break and a gradual steepening towards high frequency.

Image from ZuHone et al. (2015) of turb vs had spectra

2.3.3 Hadronic Production

For the secondary production we instead consider the protons within the ICM, rather than the electrons. We again can separate them into relativistic 'cosmic ray' protons (CRp) and thermal protons and examine their interactions. CRp have significantly longer radiative lifetimes than CRe, with $\tau_p \sim 10^{10}$ yrs, allowing them to travel up to 1 Mpc Dennison (1980). The spread out CRp p_{CR} then collide inelastically with thermal protons p_{th} to produce charged and neutral pions π

$$p_{CR} + p_{th} \rightarrow \pi^0 + \pi^+ + \pi^- \tag{2.12}$$

the charged pions then decay into (anti)muons $\mu^\pm$ and muon (anti)neutrinos $\nu_\mu(\tilde{\nu}_\mu)$, while the neutral pions decay into gamma rays γ

$$\pi^\pm \rightarrow \mu^\pm + \nu_\mu(\tilde{\nu}_\mu) \tag{2.13}$$

$$\pi^0 \rightarrow 2\gamma \tag{2.14}$$

the muons then decay into highly relativistic (positrons)electrons $e^\pm$ and electron (anti)neutrinos $\nu_e(\tilde{\nu}_e)$.

$$\mu^\pm \rightarrow e^\pm + \nu_e(\tilde{\nu}_e) + \tilde{\nu}_\mu(\nu_\mu) \tag{2.15}$$

From this chain reactions we end up with two observables, CRes and gamma rays. Importantly, it is these secondary CRe products that are predicted to produce the minihalo emission. The emission in this model then traces the population of CRp producing the CRe, rather than the CRe directly. Due to our assumption of the long radiative lifetimes, we expect a continuous injection of CRe from the hadronic process.

(ZuHone et al., 2015) investigate the hadronic method via simulation and assume a power-law momentum distribution for the CRp, $n_p(p_p) = C_p p_p^{-\alpha_p}$ (C_p is a normalisation constant), and estimate the spectral index to be in the range $\alpha_p \sim 2.0$-2.5. They then derive a distribution for the produced CRe momentum p based upon the CRp momentum distribution to be

$$Q(p)dp = \frac{4}{3} 16^{1-\alpha_p} c n_{th} C_p \sigma_{pp}(16p) p^{-\alpha_p} dp \tag{2.16}$$

with n_{th} the density of thermal protons, $\sigma_{pp}(16p)$ is the proton-proton inelastic collision cross section at 16 times the momentum.[4] Ignoring the constant terms we can also write

$$Q(p)dp \propto n_{th} \sigma_{pp} p^{-\alpha_p} dp \tag{2.17}$$

If σ_{pp} is constant[5] we are able to find a steady state for the number density of CRe by equating the synchrotron and IC losses $\frac{dp}{dt} \propto -p^2$. This occurs when $\frac{d}{dp}(-p^2 n_e) + Q(p) = 0$, which following

[4]This is an approximation of the CRp momentum in the collision, $p_p = 16p$.

[5]There is a weak dependence on the CRp and proton energies, which ZuHone et al. (2015) account for, which has a minor flattening effect on the synchrotron spectral index.

through by integrating for all particles $p \rightarrow \infty$,

$$-p^2 n_e = -\int_p^\infty Q(p)$$
$$p^2 n_e \propto \int_p^\infty p^{-\alpha_p}$$
$$n_e \propto \frac{p^{-\alpha_p+1}}{p^2}$$
$$n_e \propto p^{-(\alpha_p+1)}$$

This gives us an CRe density power-law with index $\delta = \alpha_p + 1$ and therefore a synchrotron spectral index of $\alpha = \frac{\alpha_p}{2}$. Using the expected values for α_p we then should expect $\alpha \sim$ 1-1.25 which is similar to that of observed minihalos. However, this assumes steady state injection which is an extremely idealised case for the intermittent and turbulent core of the cluster.

2.3.4 Observables of hadronic production

This idealised case predicts a flat spectral index extending into high frequencies, lacking the steepening observed in the reacceleration model. This also implies the spectral index will be constant throughout the spatial extent of the minihalo, rather than varying due to localised high amounts of turbulence. Keshet and Loeb (2010) show that changes in the magnetic field strength relative to B_{CMB} can induce changes in the emissivity. For regions of the magnetic field where $B < B_{CMB}$ they predict a radio supression $(B/B_{CMB})^2$. Assuming that the magnetic field inside a cluster has a radial steepening, there is some radius r_b where the magnetic field drops below B_{CMB} and a radial break may be observed in the radio profile.

The simulation of hadronic production by ZuHone et al. (2015) find that the majority of particles have spectral indices close to that of steady state. As B decreases towards $B \ll B_{CMB}$ the spread of spectral indices converges tightly. The opposite happens at $B \gg B_{CMB}$ with broadening and a small skew towards steeper indices. They attribute the broadening to sampling regions with rapidly decreasing magnetic fields (flatter index) and rapidly increasing magnetic fields (steeper index). They also note that the spatial bounding of the hadronic emission is less constraining than that of the reacceleration model. A drop in the emission across the cold front occurs due to a drop in the magnetic field strength and falls $\propto B^2$. As such, the extent of the minihalo is more constrained via the strength of the magnetic field, than the cold fornts of the cool core.

Importantly, ZuHone et al. (2015) do not find evidence of a spectral flattening at high frequencies, showing that a simple power-law relationship to beyond 10 GHz. They compare the spectral distribution of the hadronic and reacceleration models and clearly show the difference in spectral shape (Figure 2.2).

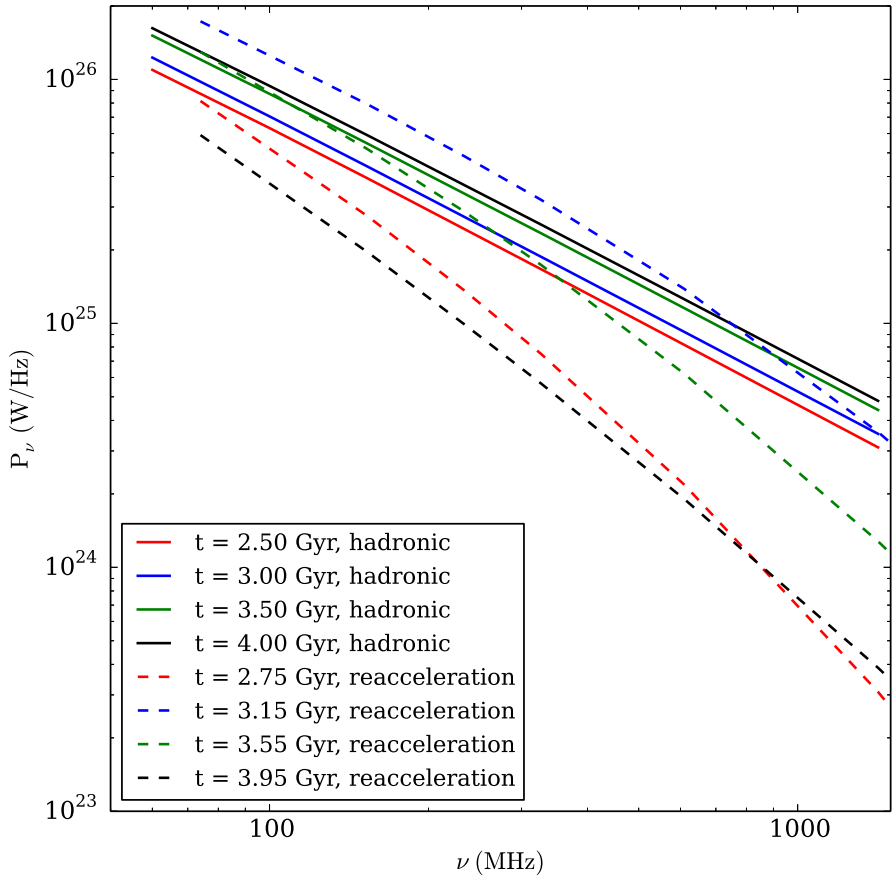

Figure 2.2: The total radio spectrum within 300 kpc radius of the cluster centre, comparing the hadronic and reacceleration model at several timesteps. (Fig 8 of ZuHone et al. 2015)

A final clear distinction from the reacceleration model is the emission of gamma rays with the hadronic model. This implies that there will be a diffuse gamma emission aligned spatially with the minihalo. This would be a 'smoking gun' proof for the hadronic model, however as mentioned earlier, current instruments don't have the spatial resolution to accurately detect diffuse gamma emission at minihalo scales.

2.4 Previous minihalo studies

2.4.1 Previous minihalo studies

- Generally look at lower frequencies, or a singular study for a particular minihalo.
- Want to look at more minihalos in a more holistic manner.
- Some findings from previous papers e.g. RXJ1720, MS1455+2232 in basic detail. (more detail on the per-halo discussion).

Chapter 3

Calibration and Imaging

This would be how to calibrate and the calibration results for the datasets, plus any special considerations we took.

3.1 What is an interferometer

- Limits of dish size in resolution, causes us to explore different ideas.
- How does an interferometer work.
- Bunch of dishes being interfered, delay to tune positioning, uv plane to determine coverage.
- From all of this we retrieve amp + phase measurements.

3.2 Our observations

- We are using VLA data in X-band, what does this mean and what are the specs for the VLA.
- These observations are of list minihalos and this setup, observation proposal.
- Original source of these minihalos is the G14 paper (among others).
- Table of minihalos + observing setup.

3.3 How to calibrate a dataset

- We have the VLA calibration pipeline to remove stress from our lives.
- We have a non-neglible amount of RFI to add stress to our lives.

- A variety of steps but first pain, what is RFI and what does it cause.

3.4 Flagging data

- Initial testing showed that the pipeline flagging steps didn't work well for our data.
- AOFLAGGER worked much better and has a mostly out-of-box VLA script.
- How does AOFLAGGER work
- Integrating into a pipeline step to still use the pipeline

3.5 Calibration pipeline

- Work through each calibration step explaining what it does and why it is important.
- Final preparation for imaging is averaging down to further reduce noise.
- Notes about general trends from the calibration, with RFI in 10-12 GHz D config being bad, much of it being flagged.

3.6 Special mentions

- Either here or during following imaging section talk about the 2-3 datasets we processed slightly differently due to bad fits.
- Describe the fitting process for the custom bandpass.

3.7 Imaging

- When we make a raw image from the data, we still have a lot of artefacts from the point spread function, which comes from the beam.
- This means we need a more advanced imaging process to remove this to get a clean image. Implemented in a few different ways, we choose tclean in CASA.
- Tclean works by modelling the PSF, creating an empty model, find brightest point in residuals, model that as real, subtract model from residuals, repeat 20K times.
- This works mostly well, but we can improve results using self calibration.

3.7.1 Self calibration

- The phase calibration is the most iffy part for the main calibration, so we can make a test image and check the model seems reasonable.
- Then use that model to make small phase corrections to our dataset. Then repeat that a few times and these corrections should converge since it is circular.
- This is the final calibration steps that we do, phase corrections 20 degrees → 1 degree after a few rounds.

3.7.2 Final cleaning

- Between self calibrations this we also check the datasets and do some final manual cleaning.
- Clipping the datasets to a sensible amp range, removing spikes, SPWs where there is small amounts of data left.
- Overview of the `assess_spw` function used for this.

Chapter 4

How to Analysis

This chapter starts with an overview on imaging and how that is performed. Then we talk a bit about how we actually want to measure the flux of the minihalos. Could combine with actual results.

4.1 Final datasets

- Could have a table here talking about each minihalo, how much data is remaining, spws that are flagged out, some sort of visual.
- Maybe bar graph of spws, line or point graph for minihalo C/D (either total or differentiated by style) normalised to [0, 1]

4.2 Calibration Error

- Calibration error exists, and sources can vary in their flux amount.
- We can try and constrain the error and variance by comparing the secondary calibrator models for D config and the point source fluxes for the images.
- Fluxes for secondaries can vary a lot over a year (100%?) maybe, but lower per month. Can then compare to differences in sources in the image.
- If a trend is observed, likely calibration error. If no trend or close models, constrain calibration error and can see the variance in AGN etc.
- Can retrieve the point source flux by cutting out diffuse sources (uvrange) and by masking to a tight border.

4.3 Calculating the minihalo flux

- We can do this in a variety of ways, but we don't want the AGN contamination so we should subtract this out.
- Do this per dataset to account for variation over time, incorrect subtractions can result in negative bowls in emission, or remaining flux.
- Other problematic sources are dealt with the same way.
- Image the subtracted data and then we create a contour for significant emission (3 sigma) from the residual RMS.
- Calculate total flux in the region.
- Then we calculate the error using this formula from G14 modified for the AGN error since we think its lower.

Chapter 5

Results

- Sections for each minihalo with a brief explanation of imaging process, custom things we had to do to get results.
- Number of self calibration, sources subtracted, difficulties in imaging certain aspects.
- Discussion on the calibration error and variance in the observed point sources.
- Includes an image of the uvrange data showing point sources, image of the minihalo with contours, and anything else we desire.
- Includes discussion compared to previous results or measured values (likely G14) and any caveats or concerns.

Chapter 6

Discussion

- General findings from the above
- Any downsides caveats for the whole dataset
- Compare general results to things like G14 / G19 and how comparable these values are. Both literally and if valid comparison.

Chapter 7

Conclusion

- I'm sure I will have some conclusions here.

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